Proposal of team software project

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Study program: Computer Science - Software and Data Engineering

Project title: SAE-Based Interpretable Neural Steering for Recommendation Systems

Project type: Research project

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1 Introduction

Recommender systems have become ubiquitous in digital platforms, yet users typically interact with them through a black box: they receive recommendations but have limited control over *how* those recommendations are generated. While users can provide implicit feedback or explicit preferences, they cannot directly influence the underlying reasoning process of the recommendation model. Although today's recommendation systems achieve impressive predictive accuracy through deep neural architectures, their internal decision processes remain opaque to end users. This lack of transparency and control leads users to feel disconnected from the recommendation process, resulting in dissatisfaction, mistrust, and reduced engagement.

Addressing this opacity requires methods that can decompose neural representations into concepts users can understand and control. Recent advances in interpretability, particularly Sparse Autoencoders (SAEs), offer such a solution by decomposing neural network representations into sparse, semantically meaningful features that are often human-interpretable. For instance, in a movie recommender, SAE features might correspond to concepts like "*1980s sci-fi*", "*strong female leads*", or "*slow-paced cinematography*". By exposing and making these features controllable, we can create a new paradigm of transparent, steerable recommendations.

1.1 Relation to other projects

This project directly extends EasyStudy[1], a framework for rapid deployment of recommender system user studies developed by the supervisor's research group at Charles University. EasyStudy provides essential infrastructure, including dataset management, preference elicitation modules, baseline algorithms, and evaluation metrics. Our work will contribute new capabilities to this framework while maintaining compatibility with its existing architecture.

Building on foundational research in mechanistic interpretability, particularly work on SAEs as decomposition tools for neural representations. This research direction, including its application to recommender systems, has been recently initiated by the project supervisor and consultants as part of the GACR 25-16785S project. Rather than developing novel SAE training methodologies, we focus on the interaction design and evaluation aspects that remain unexplored in the existing literature. Our research will complement ongoing work in explainable recommender systems and interactive machine learning, addressing a critical gap: while prior work has focused on post-hoc explanations of recommendations, we investigate whether interpretable internal representations, together with appropriate user interfaces, can serve as effective control mechanisms for preference articulation and refinement.

2 Project description

A key innovation of this project is the integration of SAE-derived feature representations directly into the user interaction loop. While we can leverage existing pipelines for feature extraction and semantic mapping, the crucial question is how to present these features in ways that users can understand and manipulate intuitively. Rather than developing novel Machine Learning (ML) architectures, we focus on the Human-Computer Interaction (HCI) challenge: designing effective paradigms that bridge the gap between SAE features and users' expectations. We will develop and implement multiple HCI concepts for feature presentation and steering input methods. Each variant will explore different assumptions about how users conceptualize recommendation criteria and how they prefer to express steering intentions. The challenge extends beyond User Interface (UI) design to understanding how users interpret features that may not map cleanly to traditional content attributes, and how to translate their steering attempts, which may be ambiguous or contradictory, into meaningful feature adjustments. These HCI experiments will establish the foundation for understanding which interaction paradigms best support user agency in neural recommendation systems.

As part of the EasyStudy integration, we will implement and evaluate multiple interface variants to assess usability and effectiveness. An additional component is extensive questionnaire support, as our task is not to conduct comprehensive comparative studies within this project, but to build the infrastructure that enables easy evaluation in follow-up research while ensuring the software components function correctly. Key variants include:

- **Steering modalities**: sliders for individual SAE features, toggles/checkboxes, example-based steering (prototype selection), and simple natural-language prompts mapped to feature weights. These variants test different cognitive models of how users prefer to express control, from precise numerical adjustment to high-level semantic guidance.

[1] https://github.com/pdokoupil/EasyStudy/tree/main

- **Presentation modes**: lexical labels with concise descriptions, prototype items per feature, and visualizations of feature activations. We explore how different representations affect feature interpretability and user confidence in steering.
- **Algorithmic options**: feature-conditioned reranking (post-hoc), latent-space perturbation with projection, and constrained optimization (soft or hard constraints). While the baseline pipeline exists, we will adapt and tune these mechanisms to ensure steering produces coherent, predictable changes in recommendations.
- **Datasets and tasks**: MovieLens[2] and GoodBooks[3] as primary domains, chosen for its interpretability (as they have well-understood attributes users can relate to) and established baseline models. We implement an abstraction layer to enable future extension to other domains (e.g., news, music), facilitating evaluation of different steering approaches and interaction paradigms.
- **Extensive questionnaire infrastructure**: case-based validation studies to ensure steering mechanisms produce expected outcomes and software components function correctly, supported by comprehensive questionnaire modules for capturing user experience, preference articulation patterns, and perceived control. The goal is to instantiate validation cases verifying steering correctness and supporting future comparative studies.

Prioritising the implementation of a few steering modalities, a basic reranker, and MovieLens experiments, we plan to extend the work based on pilot study results. Note that conducting extensive comparative user studies falls outside the project scope. Instead, we focus on developing the necessary infrastructure and validation methodology to facilitate such research in subsequent work.

Value proposition for users

Instead of merely reacting to recommendations, users gain agency over their recommendation experience. They can explore *why* items are recommended, emphasise or de-emphasise specific aspects they care about, and discover content through concept-based exploration rather than just collaborative filtering patterns. This addresses common frustrations like filter bubbles, lack of diversity, and the feeling of being "*stuck*" with certain recommendation patterns. End users thus become active participants in shaping their recommendation space, benefiting from systems that are more transparent, interpretable, and responsive to their intentions.

Value proposition for researchers

This project will produce some of the first empirical evidence of whether interpretable neural features can be effectively utilised by non-expert users in a real recommendation scenario. It will demonstrate practical applications of mechanistic interpretability beyond model debugging, potentially opening new research directions in explainable and controllable recommender systems. The extended EasyStudy framework will serve as an open-source research tool for the wider academic community, enabling reproducible experiments, comparative evaluations, and future user studies on transparency and controllability in recommender systems.

[2]https://grouplens.org/datasets/movielens/

[3]https://fastml.com/goodbooks-10k-a-new-dataset-for-book-recommendations/

2.1 Technical approach

We will build upon the existing EasyStudy architecture, adding three key components:

1. **SAE Integration Layer**: Implement interfaces for loading pretrained SAE models and extracting interpretable features from user-item interaction embeddings. This layer will handle the mapping between the recommendation model's internal representations and SAE feature activations.

2. **Feature Steering Engine**: Develop mechanisms for users to modify feature activations and propagate these changes through the recommendation pipeline. This includes both direct feature manipulation (e.g., "*increase feature #47 by 30%*") and semantic steering (e.g., "*show me more atmospheric films*").

3. **User Interface Components**: Design and implement UI elements that present interpretable features to users in accessible ways. This includes feature visualisations (what each feature represents), interactive controls for steering, and explanations connecting features to specific recommendations.

3 Evaluation methodology

Our evaluation approach prioritizes demonstrating that SAE-based steering is functionally correct and practically usable, rather than conducting comprehensive comparative studies across interface variants. The primary goal is to show that the concept works and that the interfaces are functional, not to determine which specific design is optimal.

The core of our evaluation will consist of systematic tests similar to unit tests for ML systems. We will develop case-based tests that verify steering mechanisms produce expected outcomes for well-defined scenarios. For instance, when a user increases activation of a feature associated with action-oriented content, we should observe measurable and predictable shifts in the genre distribution of recommended items. These tests will include quantitative metrics for steering fidelity how reliably user manipulations translate into the intended changes in recommendation space and will verify that different steering modalities correctly propagate user intentions through the entire pipeline.

An important aspect of our validation concerns the mapping between SAE features and user-understandable concepts. After establishing functional correctness through these tests, we will conduct limited usability testing with a small group of participants (5–10 users). The purpose is not to run a full comparative study, but rather to verify that non-expert users can understand and operate the interfaces, and to gather qualitative feedback on intuitiveness and potential improvements. This will help us identify any critical usability issues and provide preliminary evidence that the steering concept is accessible to end users.

4 Expected outcomes

The project continuation will be linked to the main branch of the EasyStudy repository[4]. During the development phase, we will determine whether to merge the current fork into the main repository or to establish a separate

[4] 1https://github.com/pdokoupil/EasyStudy/tree/main

development branch for the extended work. Since the project builds upon the existing EasyStudy framework, the deployment process will follow the same continuous integration and Docker-based pipeline already used in the repository. This ensures consistent reproducibility, maintainability, and compatibility with future system updates.

By the project's conclusion, we aim to deliver several concrete outcomes that align with our evaluation methodology. First, a fully functional extension to EasyStudy that integrates SAE-based steering capabilities, including the implementation of multiple steering interfaces and their supporting infrastructure. Each interface variant will be validated to demonstrate essential correctness in how it translates user interactions into steering operations. In simple terms, our goal is to demonstrate that the SAE-based steering approach is both technically feasible and conceptually meaningful, thereby laying the groundwork for future, more detailed user studies. Additionally, the project outcomes are intended to serve as a foundation for my subsequent diploma thesis, which will build upon and extend the results of this work.

Second, we will produce a comprehensive technical report that documents the implemented interface variants and their basic correctness. This report will present findings from our case-based testing, steering fidelity metrics across different modalities, and results from limited user validation. The report will be structured to serve either as a standalone demonstration and comparison of the implemented interfaces or as the empirical foundation for a research paper. The emphasis will be on demonstrating that each variant works as intended and on identifying which steering modality appears most generalizable, rather than declaring one approach superior to others.

Third, we will provide thorough documentation of all components, evaluation procedures, and test cases to ensure reproducibility and enable other researchers to build upon our work. This documentation will support integration of our testing framework into the broader EasyStudy ecosystem and help identify promising directions for future comprehensive user studies.

The findings and technical contributions will be documented in a form that enables future research papers. Specifically, this project will establish the foundational infrastructure and validation methodology necessary for conducting comparative studies of recommendation systems with SAE-based steering. The technical report and working system will provide the basis for subsequent papers comparing different steering approaches, interface designs, or recommendation algorithms, suitable for submission to venues or related workshops on interpretable ML. The primary contribution of this project is demonstrating the feasibility of making interpretable neural features directly controllable by end users in recommendation scenarios, thereby opening new avenues for research in transparent and steerable recommendation systems.